

Module 05: TCP Data Transfer, Pushing, and Urgent Data

Unit 4: Transport Layer

Interactive & Urgent Streams

TCP Normal Buffering Stream Model, PUSH (PSH) Immediate Transmission, URG Flag & 16-Bit Urgent Pointer Offset Math, and Out-of-Band Control Signals.

TCP Pushing Data (PSH Flag) & Urgent Data (URG Flag / Pointer)

1. Pushing Data Mechanics (PSH Flag = 1)

Default TCP Buffering Behavior:

- Sending TCP waits until buffer accumulates MSS bytes.
- Receiving TCP buffers segments until application requests read.

High delay — Unacceptable for interactive shells (Telnet / SSH)!

With PSH Flag SET (PSH = 1):

- Sending TCP flushes send buffer immediately without waiting.
- Segment transmitted across network instantly.
- Receiving TCP delivers data directly to application layer, completely bypassing receive buffer delay!

Typical Interactive Use Case:

User types a single character in SSH / Telnet session.
PSH flag forces instant echo back to terminal screen!

2. Urgent Data & Urgent Pointer (URG Flag = 1)

TCP Data Stream Buffer Layout:

Normal Data

URGENT DATA
Ctrl+C / Abort Signal

Normal Data

↑
Urgent Pointer

Urgent Pointer Calculation Formula:

End of Urgent Byte = Seq Number + Urgent Pointer - 1

- URG flag = 1 indicates 16-bit Urgent Pointer field is valid.
- Urgent data is extracted & processed out-of-order ahead of queue!

Real-World Out-of-Band Abort Case:

While downloading a massive 10 GB file via FTP, user presses **Ctrl + C (Abort command)**.
Without URG, abort command waits behind thousands of queued bytes!
With URG, server stops transmission instantly!

1. PSH vs URG Core Comparison

Feature	Pushing Data (PSH = 1)	Urgent Data (URG = 1)
Primary Goal	Eliminates buffer waiting latency	Sends out-of-band interrupt / control signals
Queue Order	Processed in-order (FIFO)	Bypasses normal queue (High-priority interrupt)

Header Field	PSH bit only	URG bit + 16-bit Urgent Pointer field
Classic Use Case	Interactive Telnet / SSH keystroke echo	User pressing Ctrl+C or FTP abort command